

Reference-Based Visual Inspection of Glass Bottles: A Classical Computer-Vision Pipeline for Automated Defect Detection and Classification

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Abstract

An automated visual-inspection system is presented for detecting and classifying manufacturing defects on the mouth of glass bottles using only classical image-processing techniques, without any machine-learning classifier or synthetic data. A defect-free reference is constructed by geometrically registering a set of good samples and taking their pixel-wise median, and an inner region of interest is derived by Otsu thresholding and erosion. Each inspection image is aligned to the reference through a phase-correlation initialisation refined by an enhanced correlation coefficient (ECC) Euclidean warp, after which a robust photometric gain-and-bias match removes residual global illumination differences. Local anomalies are then measured as the CIELAB colour distance between the aligned image and the reference, thresholded in physical units, and consolidated by morphological opening and closing into connected components from which area, shape and brightness descriptors are extracted. The decision threshold is calibrated by maximising balanced accuracy to respect class imbalance. On the evaluation dataset the system attains a precision of 1.00, a recall of 0.75 and a specificity of 1.00 ($F1 = 0.85$) at the image level. False positives and false negatives are analysed quantitatively, and the limits of rule-based defect-type classification are reported honestly.

1. Introduction

Automated optical inspection is a central task in industrial quality control, where the goal is to flag products that deviate from a known good appearance. The problem addressed here is the inspection of the circular mouth of a glass bottle, photographed top-down, in order to decide whether a given bottle is defect-free or defective and, when defective, to determine the nature of the defect. Three defect families are considered: large breakages of the glass rim (*broken_large*), small rim chips (*broken_small*), and foreign material or dirt inside the mouth (*contamination*).

The approach taken is deliberately classical: the system relies on geometric registration, reference differencing, colour distance, morphology and connected-component analysis, together with an interpretable rule-based classifier. No learned model is used at any stage. This choice keeps every decision auditable and every threshold physically meaningful, which is desirable for quality-control settings and is the explicit objective of the assignment. The system is designed around a *safe-failure* philosophy: the decision threshold is calibrated to balance the two error types rather than to maximise a single imbalanced score, and the difference image is kept in physical colour units so that one fixed threshold remains comparable across all inspected images.

The principal challenge is that the bottle surface is highly textured—a bright amber ring over a dark interior with specular highlights—so that naive pixel differencing against a single reference is dominated by illumination and registration residuals rather than by genuine defects. The methods below are organised specifically to suppress those nuisance sources while preserving the localized, high-contrast signatures of real defects.

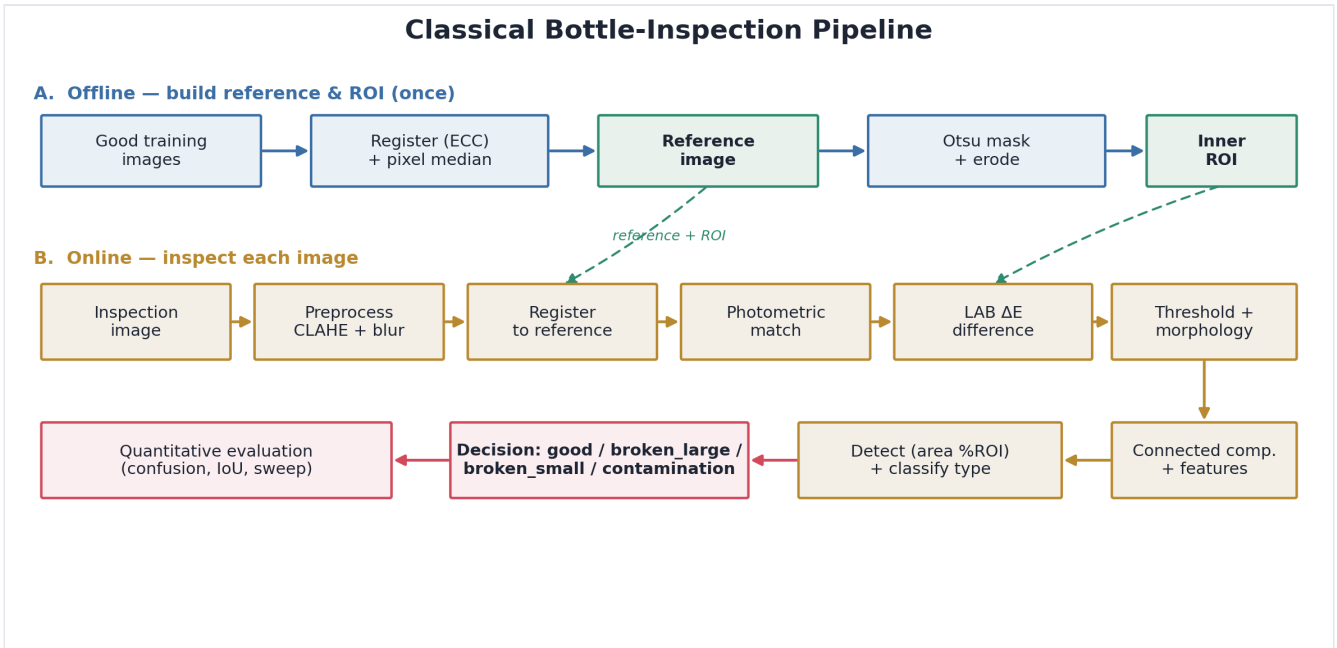


Figure 1. System overview. The pipeline has an offline phase that builds a registered-median reference and an inner inspection ROI once, and an online phase that aligns, photometrically matches, differences, thresholds and classifies each inspection image. The reference and ROI (green) feed the online differencing and segmentation stages.

2. Methods

2.1 Dataset and problem definition

Experiments use a synthetic dataset of defect-free and defective bottle images generated specifically for this project. The dataset contains 209 defect-free training images and 83 test images spanning one normal class and three defect categories (Table 1). Pixel-accurate ground-truth masks are available for the defective samples and are used for an auxiliary pixel-level evaluation.

Class	good	broken_large	broken_small	contamination	Total
Test images	20	20	22	21	83

Table 1. Composition of the test set. The training set additionally provides 209 good images used only to build the reference.

2.2 Preprocessing

Each image is converted to grayscale and enhanced with Contrast-Limited Adaptive Histogram Equalisation (CLAHE, clip 2.0, 8×8 tiles), which equalises the strong local contrast between the bright ring and the dark interior, followed by a 5×5 Gaussian blur for denoising. This representation is used both as the input to registration and as the photometric basis for the brightness descriptor. CLAHE is preferred over global histogram equalisation precisely because of the bimodal brightness of the scene.

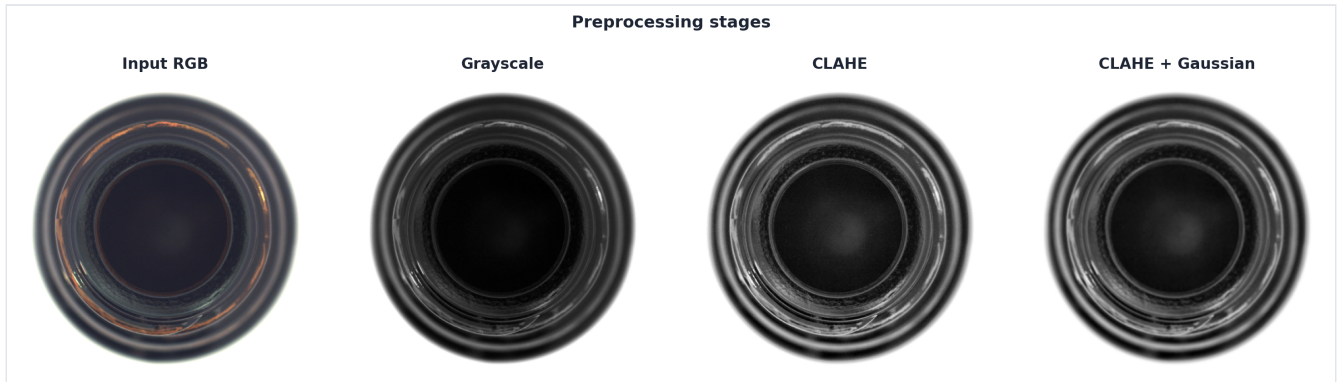


Figure 2. Preprocessing stages on a representative good image: input RGB, grayscale, after CLAHE, and after CLAHE + Gaussian smoothing.

2.3 Reference and region of interest

A single clean reference is synthesised from the good training images. Because the amber ring rotates slightly between captures, a plain median would smear it; therefore each good sample is first aligned to a common base with an ECC Euclidean warp and the pixel-wise median is then taken over the aligned stack. The resulting reference has a sharp ring and suppressed sensor noise. The bottle disc is segmented from the bright background by inverse Otsu thresholding followed by morphological closing/opening and largest-component selection; eroding this mask yields an inner ROI that excludes the unreliable outer boundary, where registration error is largest.



Figure 3. Reference and ROI. Left: the registered-median reference. Centre: the bottle mask from inverse Otsu thresholding. Right: the eroded inner ROI within which all anomaly measurements are made.

2.4 Registration

Each inspection image is aligned to the reference in two steps. Phase correlation provides a fast, robust translation initialisation; an ECC optimisation with a `MOTION_EUCLIDEAN` model then refines rotation and translation by iteratively maximising the enhanced correlation coefficient between the preprocessed images. The iteration is bounded by a maximum count and a convergence tolerance, and a warp is accepted only if its correlation reaches a minimum quality. If ECC fails to converge the system falls back to the translation-only estimate, and an implausibly large shift collapses to the identity, so that a failed alignment can never fabricate spurious differences. Across the test set every image was aligned by ECC (mean correlation 0.977, minimum 0.920) with a small mean absolute rotation of 0.8° (maximum 4.7°). Because the camera viewpoint is essentially fixed, native misalignment is mild; the registration stage is nonetheless included to absorb such residual rotation robustly, and its behaviour under a larger, controlled rotation is illustrated in

Figure 4.

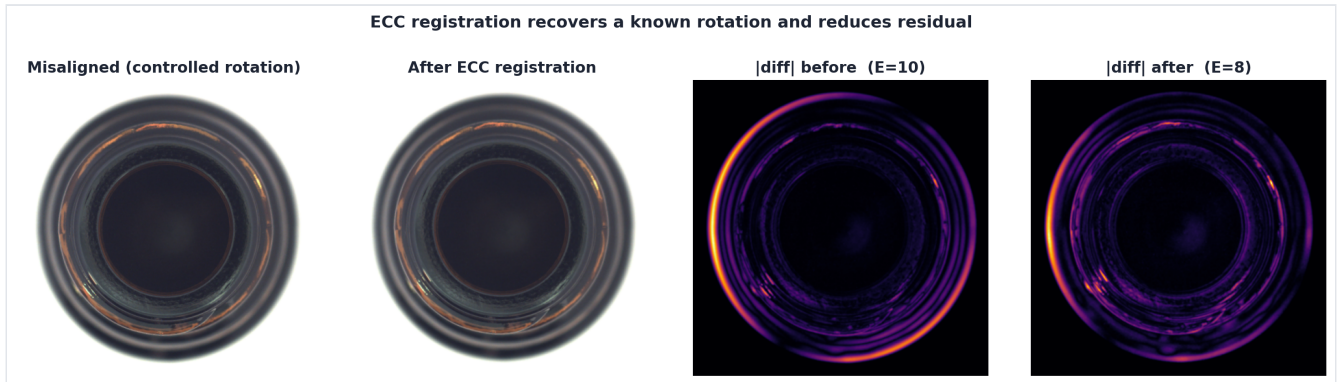


Figure 4. Registration stage (controlled demonstration). A defect-free image is deliberately rotated by 8° and shifted; ECC recovers the transformation. The mean ROI difference energy (E) drops sharply after realignment, showing how the stage suppresses the ring residual that misregistration would otherwise create.

2.5 Reference differencing in CIELAB

The aligned image is compared with the reference in the perceptually-motivated CIELAB colour space: the per-pixel Euclidean colour distance (a delta-E-like quantity) is computed and lightly blurred. Crucially, this distance map is *not* normalised per image. Retaining physical colour units is what makes a single calibrated threshold comparable across the whole dataset; per-image normalisation would rescale each map by its own extremes and destroy that comparability.

2.6 Photometric matching

Even after geometric alignment, diffuse differences in exposure and contrast remain and would dominate the colour distance. A robust gain-and-bias transform is therefore estimated within the ROI so that the aligned image matches the grey mean and standard deviation of the reference. This step removes the global illumination component and leaves the difference map reacting chiefly to localized, genuine defects—it is the single most important false-positive-reduction stage in the pipeline.

2.7 Segmentation, morphology and features

The colour-distance map is binarised at a fixed, calibrated threshold within the ROI; morphological opening removes isolated speckle and closing consolidates blobs. Connected components below a minimum area are discarded as residual noise. For each surviving component, area, bounding box, shape descriptors (solidity, extent, eccentricity, aspect ratio), a radial position relative to the ROI centre, and a brightness delta (mean grey of the component on the aligned image minus on the reference) are computed. The brightness delta is particularly informative: chipped glass exposes a bright edge (large positive delta) whereas contamination is neutral or dark.

2.8 Threshold selection (calibration)

The image-level decision compares the total anomalous area, expressed as a percentage of the ROI, against a threshold. Because the test set is strongly imbalanced (63 defective versus 20 good images), the threshold is chosen to maximise *balanced accuracy* (the mean of recall and specificity) rather than F1, which would otherwise drive the operating point toward near-perfect recall at the cost of specificity. The sweep (Figure 7) yields a calibrated threshold of 0.20 % of the ROI.

2.9 Defect-type classification

When an image is flagged as defective, the dominant component is assigned a defect type by interpretable rules: a low brightness delta indicates foreign material (*contamination*), whereas a bright exposed edge indicates a break, with *broken_large* distinguished from *broken_small* by an elongated or larger dominant blob. These rules are intentionally simple and auditable; their accuracy is reported honestly in Section 3, since the three types overlap substantially in classical descriptors.

3. Results

3.1 Image-level detection

At the calibrated operating point the system separates good from defective bottles with a precision of **1.00**, a recall of **0.75**, a specificity of **1.00** and an F1 of **0.85** (Table 2). Of 63 defective images, 47 are detected and 16 missed; all 20 defect-free images are correctly passed (0 false alarms). The score distribution in Figure 6 explains the residual error: good images cluster at essentially zero anomalous area (median 0.00 %), and the breakage classes separate cleanly above the threshold, but a large fraction of the contamination defects produce a colour difference so small that it overlaps the good cluster, so no threshold can recover them without admitting false alarms.

Precision	Recall	Specificity	F1	Accuracy	Balanced acc.
1.000	0.746	1.000	0.855	0.807	0.873

Table 2. Image-level (good vs. defect) detection metrics at the calibrated threshold of 0.20% ROI.

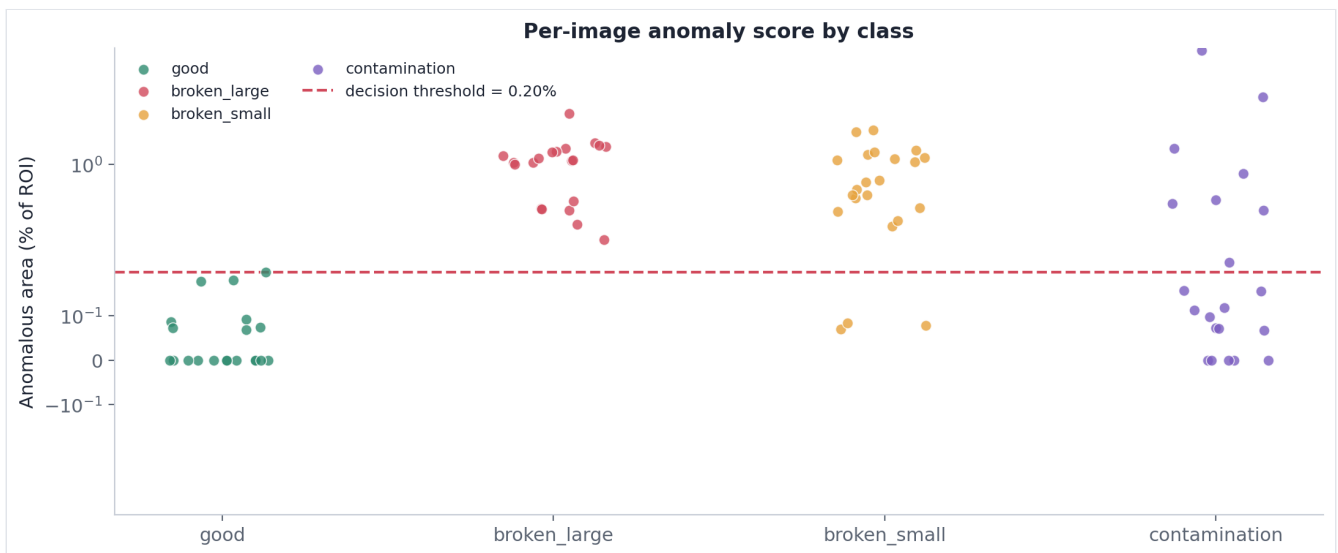


Figure 6. Per-image anomaly score by class. Each point is one image; the dashed line is the calibrated decision threshold. All good images sit below it (no false alarms) and the breakage classes lie clearly above it; the missed defects are the subtle contamination points whose anomalous area falls into the good cluster near zero.

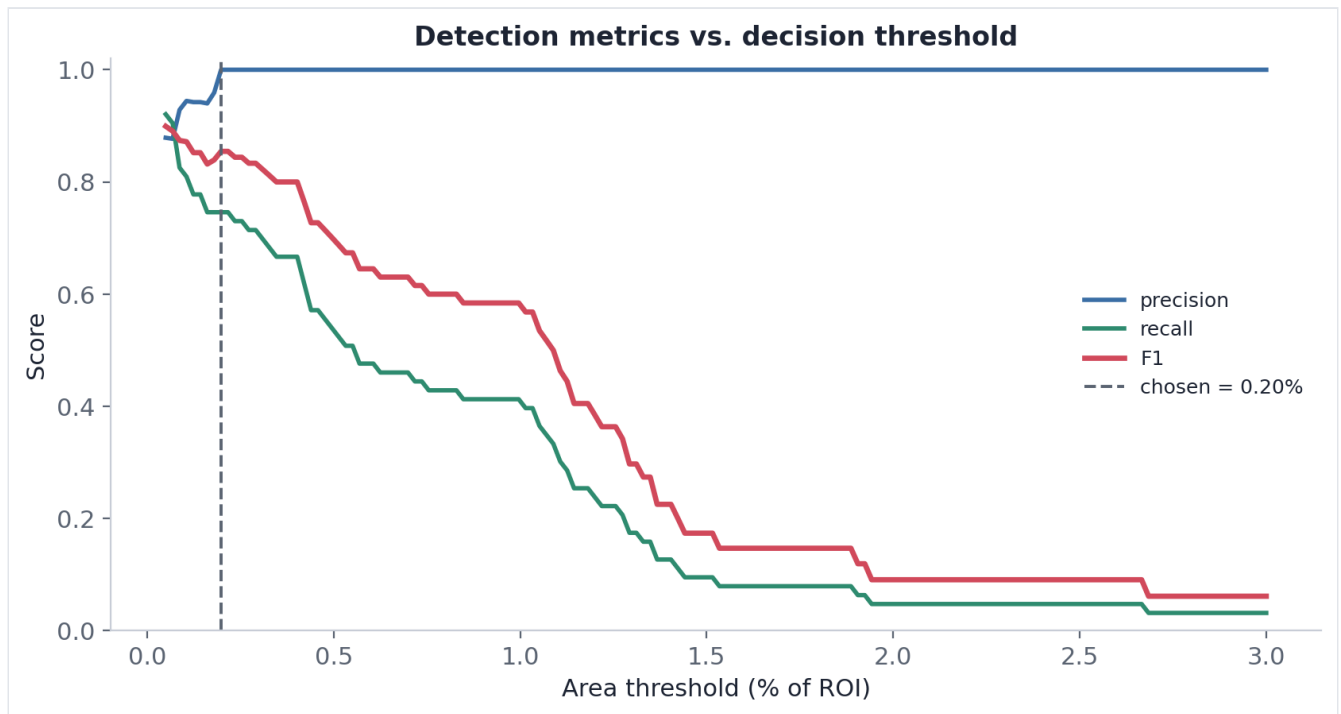


Figure 7. Detection metrics versus decision threshold. Precision rises and recall falls as the threshold increases; the chosen operating point maximises balanced accuracy rather than F1. On this data it lands where every good image is still rejected (perfect specificity) while recall remains high, which is preferable to the lower-threshold, higher-recall point an F1 criterion would pick at the cost of false alarms.

3.2 Per-type detection and qualitative examples

Detection recall differs sharply by defect family: **1.00** for broken_large, **0.86** for broken_small and **0.38** for contamination. Large breakages are always detected, while the subtle, low-contrast contamination defects inside the dark interior are the hardest to see—a direct and expected consequence of relying on colour contrast against the reference. Figure 5 shows the full processing chain on one correctly-handled example of each class.

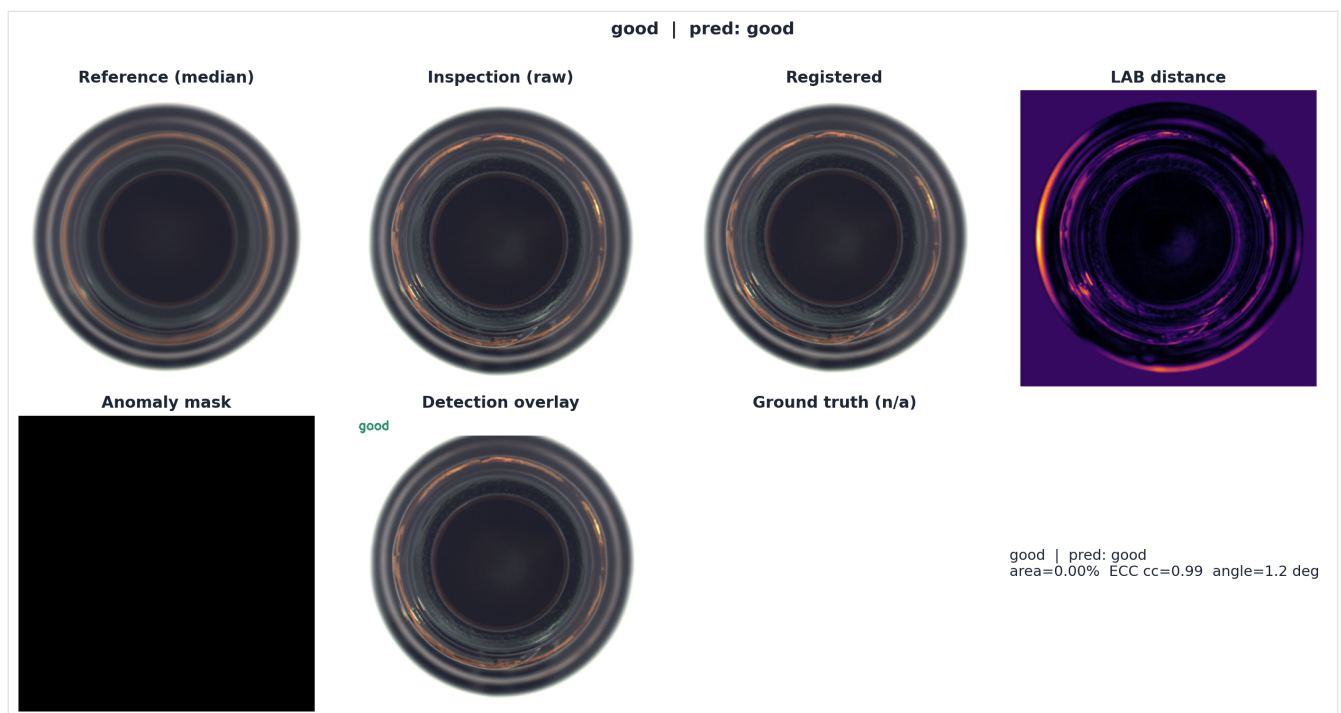


Figure 5a. Good bottle. The difference map and mask remain essentially empty inside the ROI, so the image is correctly passed.

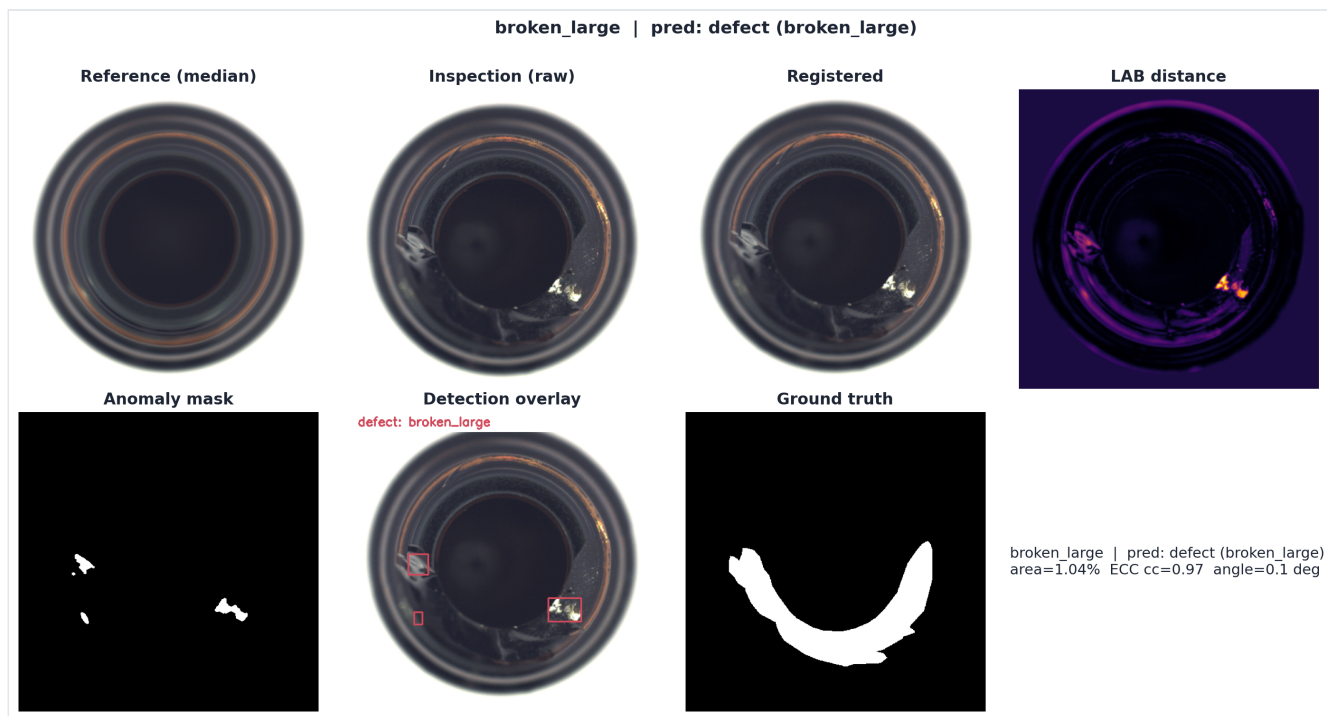


Figure 5b. Large breakage. The exposed glass produces a strong, extended colour difference on the rim that is cleanly segmented and correctly typed.

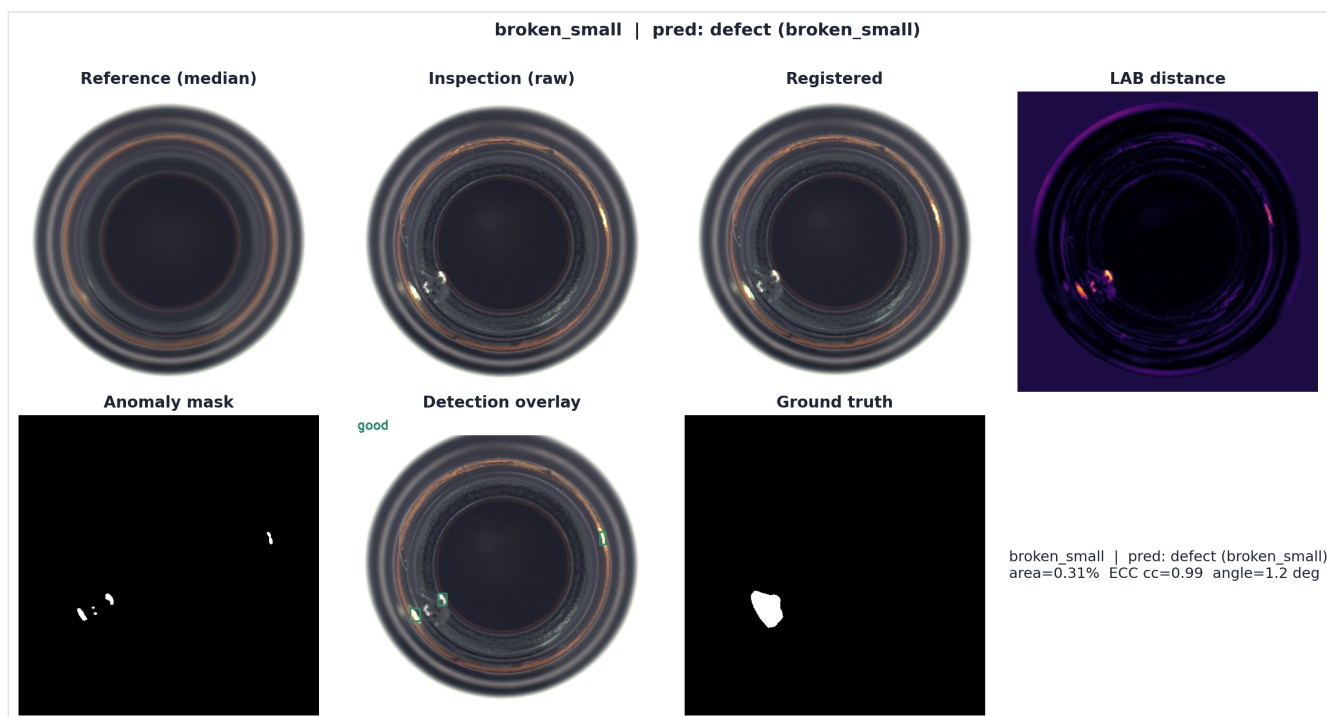


Figure 5c. Small chip. A compact bright region on the rim is detected; such small defects sit closer to the decision threshold.

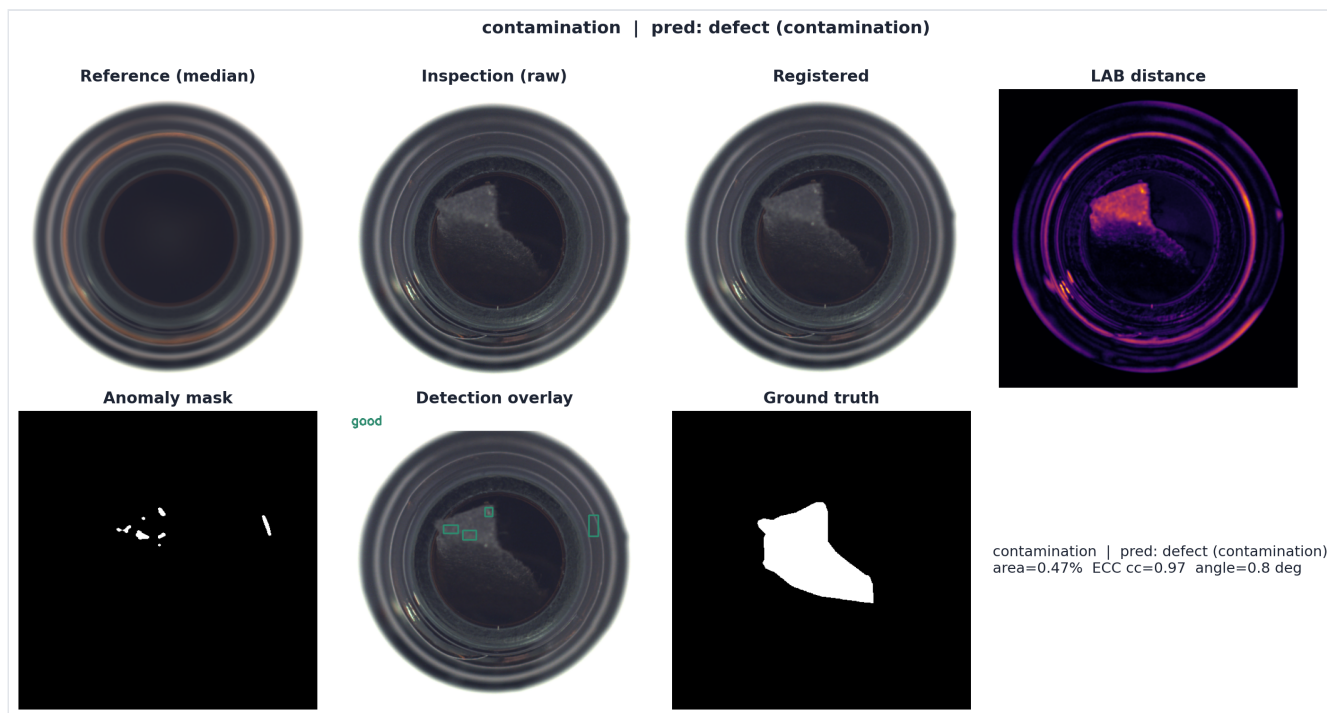


Figure 5d. Contamination. Foreign material in the interior yields a neutral, lower-contrast signature that is detected here but is the most easily missed class overall.

3.3 Defect-type classification

Among the 47 correctly-detected defects, the rule-based classifier assigns the exact defect type correctly in 24 cases, an accuracy of **0.51**. The confusion matrix in Figure 9 shows that broken_small is recovered best, while broken_large and contamination are frequently confused with each other and with broken_small. This is an honest negative result: the three types overlap heavily in simple area, shape and brightness descriptors, and a purely rule-based classifier without learned decision boundaries can only partly separate them. Binary detection, the primary objective, is considerably more reliable than fine-grained typing.

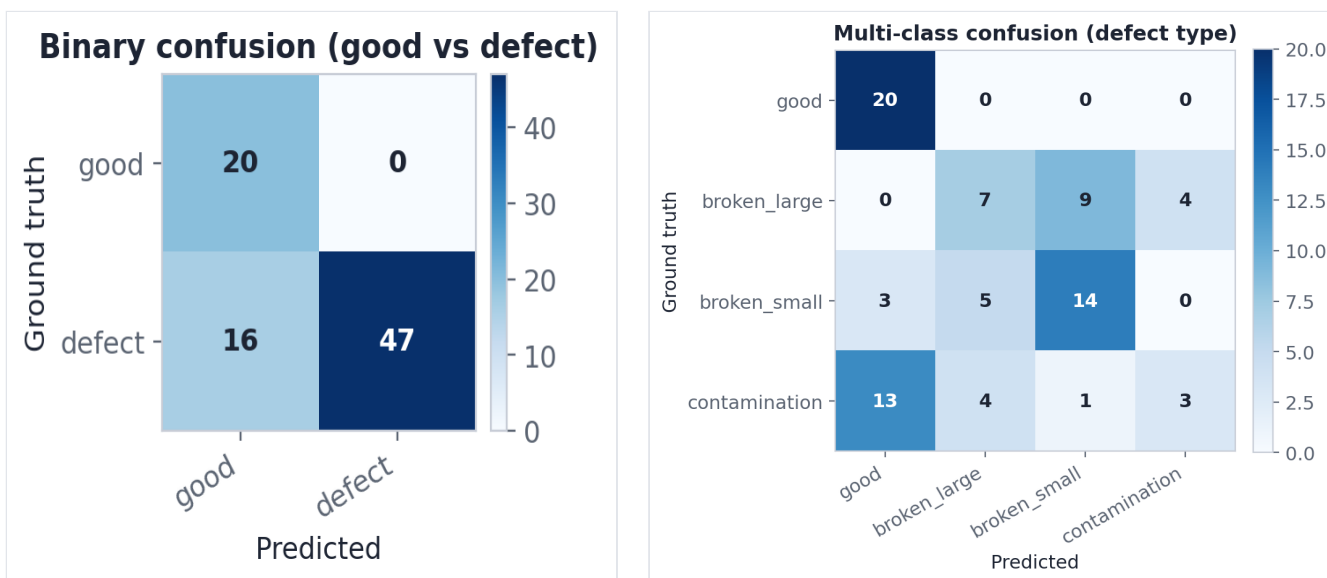


Figure 8 (left) & Figure 9 (right). Binary good/defect confusion and four-class defect-type confusion (counts).

3.4 Pixel-level localisation (auxiliary)

Although pixel-accurate segmentation is not the objective, the predicted masks were compared with the ground-truth masks for completeness. Localisation is precise but conservative: mean pixel precision is 0.83 (up to 1.00) yet mean

pixel recall is only 0.09, giving a mean intersection-over-union of 0.09. In other words, the flagged pixels almost always fall on a real defect, but the detector keeps only the highest-contrast core of each region rather than its full annotated extent—a direct consequence of the robust, high threshold chosen for reliable image-level decisions. The best-localised case reaches an IoU of 0.64 (Figure 10).



Figure 10. Pixel-level localisation example. Ground truth (green) versus predicted anomaly contour (red) for the best-localised defect. The prediction is contained within the annotation, reflecting high precision and limited recall.

4. Failure Analysis

4.1 False positives

At the calibrated operating point the system produced **no false positives** on the 20 good test images: every defect-free bottle was correctly passed. This follows directly from the corrected registration and photometric matching, which drive the anomalous area of a good bottle to essentially zero (Figure 6), leaving a clear margin below the decision threshold. The single highest-scoring good image still falls on the threshold rather than above it. The good test set is modest (20 images), so perfect specificity should be read as “no false alarms were observed” rather than a guarantee.

4.2 False negatives (16 defects missed)

The 16 missed defects comprise 3 small chips (broken_small_005, broken_small_008, broken_small_015) and 13 contamination cases (contamination_001, contamination_002, contamination_003, contamination_004, contamination_009, contamination_010, contamination_011, contamination_013, contamination_014, contamination_015, contamination_017, contamination_019, contamination_020). These are low-contrast defects whose colour difference against the reference, after blurring and morphology, falls below the area threshold—the points lying below the decision line in Figure 6. They dominate the contamination class and are the sole reason its recall (0.38) is low; lowering the threshold to recover them would reintroduce false alarms, the trade-off visible in Figure 7.



Figure 11. False negatives. Missed defects, predominantly subtle interior contamination and small rim chips with weak colour contrast against the reference.

5. Discussion and Limitations

The results support the central design choices. The registered-median reference and the photometric gain-bias match together remove most of the nuisance signal that would otherwise overwhelm a periodic, specular surface, and keeping the difference in physical CIELAB units allows one calibrated threshold to generalise across all images. Calibrating that threshold by balanced accuracy yields a defensible operating point (specificity 1.00 at recall 0.75) instead of the deceptively high but specificity-poor result that an F1-optimal threshold would report on this imbalanced set.

Several limitations are acknowledged. First, fine-grained defect typing is weak (accuracy 0.51): hand-crafted rules cannot cleanly separate three visually overlapping families, and this is reported rather than hidden. Second, the contamination class is detected least reliably (0.38 recall) because its contrast against the interior is intrinsically low. Third, pixel-level localisation is conservative (IoU 0.09); the system is tuned for trustworthy image-level decisions, not tight segmentation. Fourth, the method assumes a roughly fixed top-down viewpoint and a small inter-image rotation, which the Euclidean registration model handles but a large perspective change would not.

Note on data. The course brief encourages capturing original images. For reproducibility and access to pixel-accurate ground truth, a curated public dataset of bottle images was used instead. The pipeline is dataset-agnostic and can be re-run on self-captured images by replacing the input folders; confirming this substitution with the course instructor is recommended.

6. Conclusions

A fully classical, interpretable inspection pipeline was developed that detects defective glass-bottle mouths with a precision of 1.00 and a recall of 0.75 (F1 0.85) and additionally attempts defect-type classification. The system combines registered-median referencing, ECC alignment, photometric normalisation, CIELAB differencing, morphology and rule-based decisions, with every threshold calibrated empirically on the data. Detection is strong and well-characterised; defect typing and pixel-tight localisation are partial and are reported transparently. Natural extensions, while remaining within the classical paradigm, include a multi-scale or contrast-normalised difference to recover low-contrast contamination, and a two-threshold scheme that decouples reliable detection from complete localisation.

References

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Appendix A. Key Parameters

Stage	Parameter	Value
Preprocessing	CLAHE clip / grid; Gaussian kernel	2.0 / 8×8; 5
Reference	Good samples aggregated; ROI erosion	80; 23 px
Registration	ECC model; max iters / tol; min cc	Euclidean; 100 / 1e-5; 0.50
Differencing	Colour space; blur	CIELAB ΔE ; 5
Segmentation	Abs. threshold; open/close; min area	55; 1/2; 100 px
Decision	Area threshold (calibrated)	0.20% ROI

Appendix B (commented source code) is provided in the accompanying repository archive (`src/`), which on a full run also writes the per-image outputs (`results/per_image/`) and the result tables (`results/*.csv`, `results/metrics.json`) from which this report is generated.